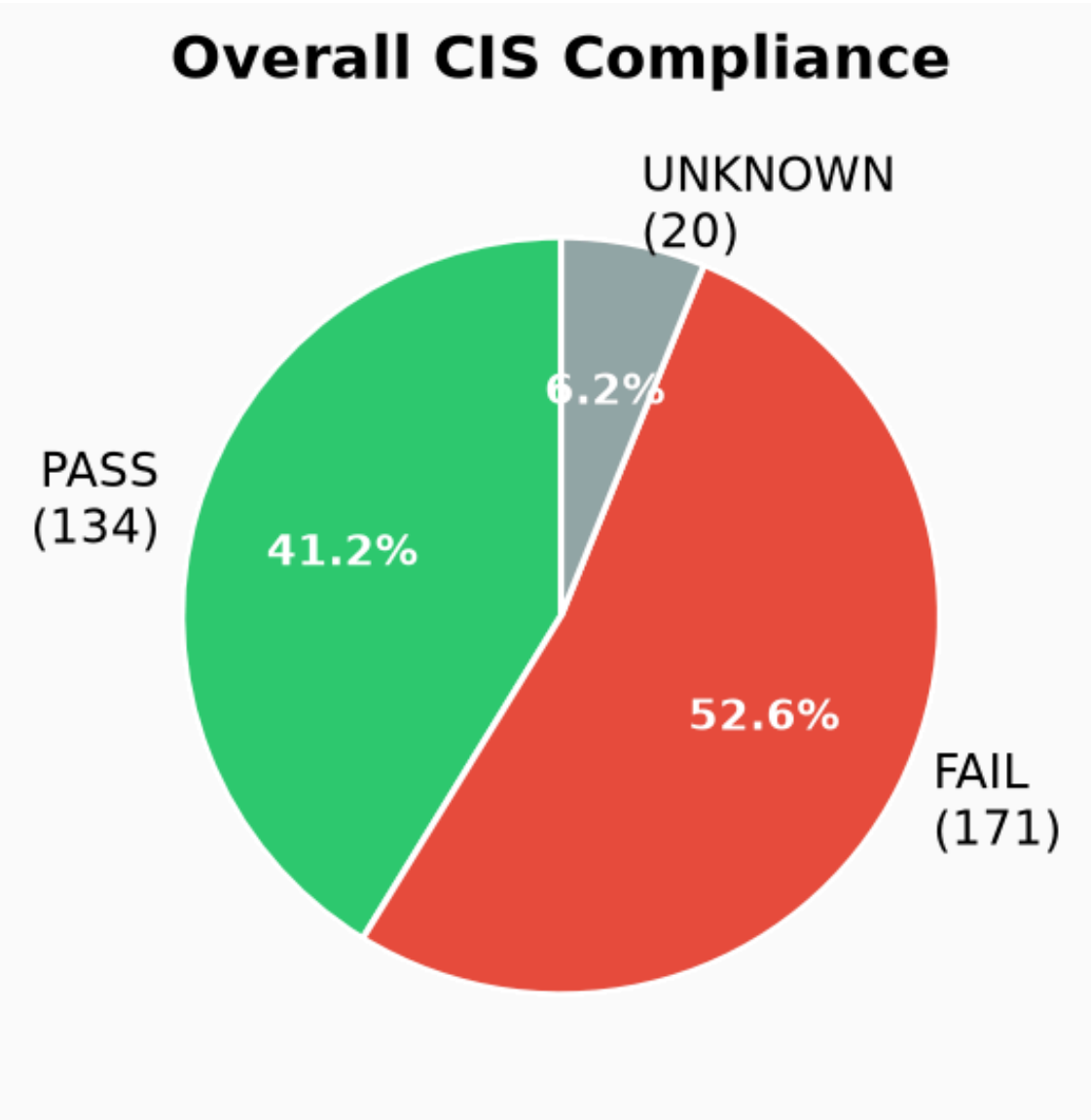


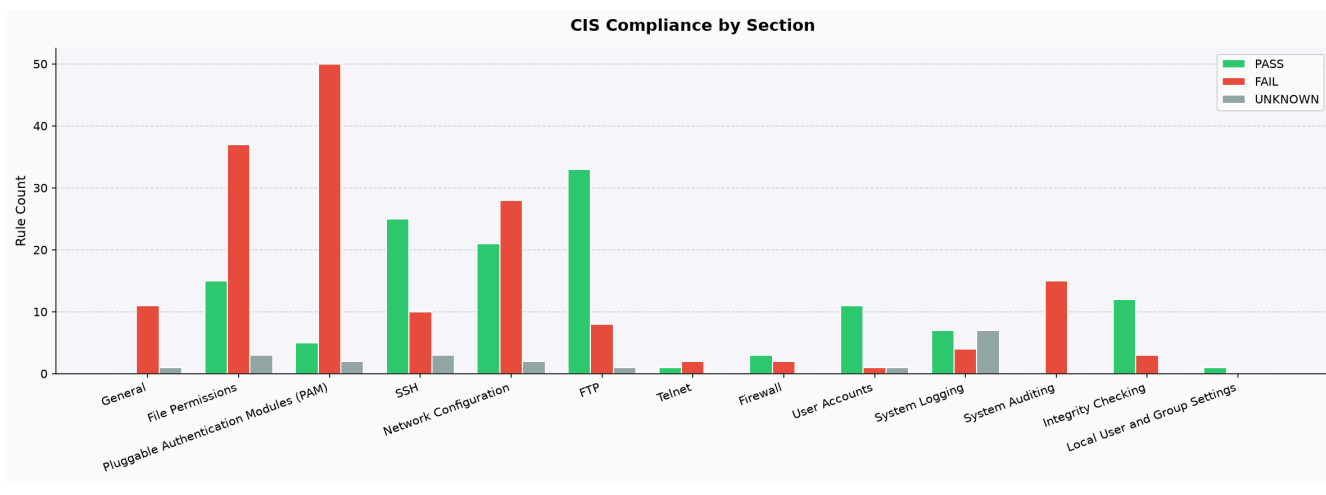
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	325
✓ Passed	134
✗ Failed	171
? Unknown	20
Overall Pass Rate	41.2%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
General	0	11	1	0%
File Permissions	15	37	3	27%
Pluggable Authentication Modules (PAM)	5	50	2	9%
SSH	25	10	3	66%
Network Configuration	21	28	2	41%
FTP	33	8	1	79%
Telnet	1	2	0	33%
Firewall	3	2	0	60%
User Accounts	11	1	1	85%
System Logging	7	4	7	39%
System Auditing	0	15	0	0%
Integrity Checking	12	3	0	80%
Local User and Group Settings	1	0	0	100%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and security profile.

Section 1.1 – Filesystem Kernel Modules & Partitions

Rule 1.1.1.1: Ensure cramfs kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `cramfs` module is not currently loaded.
 - `modprobe_config_lines` for `cramfs` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install cramfs /bin/false` to a file like `/etc/modprobe.d/cramfs.conf`.
- Add `blacklist cramfs` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.2: Ensure freevxfs kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `freevxfs` module is not currently loaded.
- `modprobe_config_lines` for `freevxfs` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install freevxfs /bin/false` to a file like `/etc/modprobe.d/freevxfs.conf`.
- Add `blacklist freevxfs` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.3: Ensure hfs kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `hfs` module is not currently loaded.
- `modprobe_config_lines` for `hfs` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install hfs /bin/false` to a file like `/etc/modprobe.d/hfs.conf`.
- Add `blacklist hfs` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.4: Ensure hfsplus kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `hfsplus` module is not currently loaded.
- `modprobe_config_lines` for `hfsplus` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install hfsplus /bin/false` to a file like `/etc/modprobe.d/hfsplus.conf`.
- Add `blacklist hfsplus` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.5: Ensure jffs2 kernel module is not available

- **Status:** FAIL

- **Evidence:**

- ☐ jffs2 module is not currently loaded.
- ☐ modprobe_config_lines for jffs2 is empty, indicating no explicit install /bin/false or blacklist configuration.

- **Recommendation:**

- ☐ Add `install jffs2 /bin/false` to a file like `/etc/modprobe.d/jffs2.conf`.
- ☐ Add `blacklist jffs2` to the same file.
- ☐ Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.6: Ensure overlay kernel module is not available

- **Status:** FAIL

- **Evidence:**

- ☐ overlay module is not currently loaded.
- ☐ modprobe_config_lines for overlay is empty, indicating no explicit install /bin/false or blacklist configuration.

- **Recommendation:**

- ☐ Add `install overlay /bin/false` to a file like `/etc/modprobe.d/overlay.conf`.
- ☐ Add `blacklist overlay` to the same file.
- ☐ Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.7: Ensure squashfs kernel module is not available

- **Status:** FAIL

- **Evidence:**

- ☐ squashfs module is not currently loaded.
- ☐ modprobe_config_lines for squashfs is empty, indicating no explicit install /bin/false or blacklist configuration.

- **Recommendation:**

- ☐ Add `install squashfs /bin/false` to a file like `/etc/modprobe.d/squashfs.conf`.
- ☐ Add `blacklist squashfs` to the same file.
- ☐ Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.8: Ensure udf kernel module is not available

- **Status:** FAIL

- **Evidence:**

- ☐ udf module is not currently loaded.

- `modprobe_config_lines` for `udf` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install udf /bin/false` to a file like `/etc/modprobe.d/udf.conf`.
- Add `blacklist udf` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.9: Ensure firewire-core kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `firewire-core` module is not currently loaded.
- `modprobe_config_lines` for `firewire-core` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install firewire-core /bin/false` to a file like `/etc/modprobe.d/firewire-core.conf`.
- Add `blacklist firewire-core` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.10: Ensure usb-storage kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `usb-storage` module is not currently loaded.
- `modprobe_config_lines` for `usb-storage` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install usb-storage /bin/false` to a file like `/etc/modprobe.d/usb-storage.conf`.
- Add `blacklist usb-storage` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 1.1.1.11: Ensure unused filesystems kernel modules are not available

- **Status:** INFORMATIONAL

- **Evidence:** The following kernel modules are currently loaded: `tcp_diag`, `udp_diag`, `inet_diag`, `nf_tables`, `tls`, `qrtr`, `cfg80211`, `binfmt_misc`, `snd_intel8x0`, `intel_rapl_msr`, `snd_ac97_codec`, `ac97_bus`, `snd_pcm`, `intel_rapl_common`, `vmwgfx`, `snd_timer`, `drm_ttm_helper`, `snd`, `i2c_piix4`, `vboxguest`, `soundcore`, `ttm`, `joydev`, `input_leds`, `mac_hid`, `serio_raw`, `sch_fq_codel`, `dm_multipath`, `msr`, `efi_pstore`, `nfnetlink`, `dmi_sysfs`, `ip_tables`, `x_tables`, `autofs4`, `btrfs`, `blake2b_generic`, `raid10`, `raid456`, `async_raid6_recov`, `async_memcpy`, `async_pq`, `async_xor`, `async_tx`, `xor`, `raid6_pq`, `libcrc32c`, `raid1`, `raid0`, `crct10dif_pclmul`, `crc32_pclmul`, `polyval_clmulni`, `polyval_generic`, `ghash_clmulni_intel`, `sha256_ssse3`,

sha1_ssse3, vga16fb, vgastate, psmouse, ahci, e1000, libahci, pata_acpi, video, wmi, hid_generic, usbhid, hid, aesni_intel, crypto_simd, cryptd.

- **Recommendation:** None. Review the list of loaded modules and disable any that are not required for system operation.

Rule 1.1.2.1.1: Ensure /tmp is tmpfs or a separate partition

- **Status:** FAIL
- **Evidence:** The /tmp mount point is not a separate partition (is_separate_partition: false) and its filesystem type is ext4 (fstype: ext4), not tmpfs.
- **Recommendation:** Create a separate partition for /tmp or configure /tmp as a tmpfs filesystem. For tmpfs, add tmpfs /tmp tmpfs defaults,rw,nosuid,nodev,noexec,relatime 0 0 to /etc/fstab and run `sudo mount -o remount /tmp`.

Rule 1.1.2.1.2: Ensure nodev option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The /tmp mount point does not have the nodev option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nodev option to the /tmp entry in /etc/fstab and remount the partition. Example: `UUID=<UUID> /tmp ext4 defaults,nodev 0 2` or `tmpfs /tmp tmpfs defaults,rw,nosuid,nodev,noexec,relatime 0 0`.

Rule 1.1.2.1.3: Ensure nosuid option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The /tmp mount point does not have the nosuid option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nosuid option to the /tmp entry in /etc/fstab and remount the partition. Example: `UUID=<UUID> /tmp ext4 defaults,nosuid 0 2` or `tmpfs /tmp tmpfs defaults,rw,nosuid,nodev,noexec,relatime 0 0`.

Rule 1.1.2.1.4: Ensure noexec option set on /tmp partition

- **Status:** FAIL
- **Evidence:** The /tmp mount point does not have the noexec option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the noexec option to the /tmp entry in /etc/fstab and remount the partition. Example: `UUID=<UUID> /tmp ext4 defaults,noexec 0 2` or `tmpfs /tmp tmpfs defaults,rw,nosuid,nodev,noexec,relatime 0 0`.

Rule 1.1.2.2.1: Ensure /dev/shm is tmpfs or a separate partition

- **Status:** PASS
- **Evidence:** The /dev/shm mount point is a separate partition (is_separate_partition: true) and its filesystem type is tmpfs (fstype: tmpfs).
- **Recommendation:** None.

Rule 1.1.2.2.2: Ensure nodev option set on /dev/shm partition

- **Status:** PASS
- **Evidence:** The /dev/shm mount point has the nodev option set (mount_options_list: ["rw", "nosuid", "nodev", "inode64"]).
- **Recommendation:** None.

Rule 1.1.2.2.3: Ensure nosuid option set on /dev/shm partition

- **Status:** PASS
- **Evidence:** The /dev/shm mount point has the nosuid option set (mount_options_list: ["rw", "nosuid", "nodev", "inode64"]).
- **Recommendation:** None.

Rule 1.1.2.2.4: Ensure noexec option set on /dev/shm partition

- **Status:** FAIL
- **Evidence:** The /dev/shm mount point does not have the noexec option set (mount_options_list: ["rw", "nosuid", "nodev", "inode64"]).
- **Recommendation:** Add the noexec option to the /dev/shm entry in /etc/fstab and remount the partition. Example: tmpfs /dev/shm tmpfs defaults,rw,nosuid,nodev,noexec 0 0.

Rule 1.1.2.3.1: Ensure separate partition exists for /home

- **Status:** FAIL
- **Evidence:** The /home mount point is not a separate partition (is_separate_partition: false).
- **Recommendation:** Create a separate partition for /home to isolate user data from the operating system.

Rule 1.1.2.3.2: Ensure nodev option set on /home partition

- **Status:** FAIL
- **Evidence:** The /home mount point does not have the nodev option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nodev option to the /home entry in /etc/fstab and remount the partition. Example: UUID=<UUID> /home ext4 defaults,nodev 0 2.

Rule 1.1.2.3.3: Ensure nosuid option set on /home partition

- **Status:** FAIL
- **Evidence:** The /home mount point does not have the nosuid option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nosuid option to the /home entry in /etc/fstab and remount the partition. Example: UUID=<UUID> /home ext4 defaults,nosuid 0 2.

Rule 1.1.2.4.1: Ensure separate partition exists for /var

- **Status:** FAIL
- **Evidence:** The /var mount point is not a separate partition (is_separate_partition: false).

- **Recommendation:** Create a separate partition for /var to prevent log files and other variable data from filling up the root filesystem.

Rule 1.1.2.4.2: Ensure nodev option set on /var partition

- **Status:** FAIL
- **Evidence:** The /var mount point does not have the nodev option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nodev option to the /var entry in /etc/fstab and remount the partition. Example: UUID=<UUID> /var ext4 defaults,nodev 0 2.

Rule 1.1.2.4.3: Ensure nosuid option set on /var partition

- **Status:** FAIL
- **Evidence:** The /var mount point does not have the nosuid option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nosuid option to the /var entry in /etc/fstab and remount the partition. Example: UUID=<UUID> /var ext4 defaults,nosuid 0 2.

Rule 1.1.2.5.1: Ensure separate partition exists for /var/tmp

- **Status:** FAIL
- **Evidence:** The /var/tmp mount point is not a separate partition (is_separate_partition: false).
- **Recommendation:** Create a separate partition for /var/tmp to isolate temporary files from other /var data.

Rule 1.1.2.5.2: Ensure nodev option set on /var/tmp partition

- **Status:** FAIL
- **Evidence:** The /var/tmp mount point does not have the nodev option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nodev option to the /var/tmp entry in /etc/fstab and remount the partition. Example: UUID=<UUID> /var/tmp ext4 defaults,nodev 0 2.

Rule 1.1.2.5.3: Ensure nosuid option set on /var/tmp partition

- **Status:** FAIL
- **Evidence:** The /var/tmp mount point does not have the nosuid option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the nosuid option to the /var/tmp entry in /etc/fstab and remount the partition. Example: UUID=<UUID> /var/tmp ext4 defaults,nosuid 0 2.

Rule 1.1.2.5.4: Ensure noexec option set on /var/tmp partition

- **Status:** FAIL
- **Evidence:** The /var/tmp mount point does not have the noexec option set (mount_options_list: ["rw", "relatime"]).
- **Recommendation:** Add the noexec option to the /var/tmp entry in /etc/fstab

and remount the partition. Example: `UUID=<UUID> /var/tmp ext4 defaults,noexec 0 2.`

Rule 1.1.2.6.1: Ensure separate partition exists for /var/log

- **Status:** FAIL
- **Evidence:** The /var/log mount point is not a separate partition (`is_separate_partition: false`).
- **Recommendation:** Create a separate partition for /var/log to prevent log files from filling up the root filesystem.

Rule 1.1.2.6.2: Ensure nodev option set on /var/log partition

- **Status:** FAIL
- **Evidence:** The /var/log mount point does not have the nodev option set (`mount_options_list: ["rw", "relatime"]`).
- **Recommendation:** Add the nodev option to the /var/log entry in /etc/fstab and remount the partition. Example: `UUID=<UUID> /var/log ext4 defaults,nodev 0 2.`

Rule 1.1.2.6.3: Ensure nosuid option set on /var/log partition

- **Status:** FAIL
- **Evidence:** The /var/log mount point does not have the nosuid option set (`mount_options_list: ["rw", "relatime"]`).
- **Recommendation:** Add the nosuid option to the /var/log entry in /etc/fstab and remount the partition. Example: `UUID=<UUID> /var/log ext4 defaults,nosuid 0 2.`

Rule 1.1.2.6.4: Ensure noexec option set on /var/log partition

- **Status:** FAIL
- **Evidence:** The /var/log mount point does not have the noexec option set (`mount_options_list: ["rw", "relatime"]`).
- **Recommendation:** Add the noexec option to the /var/log entry in /etc/fstab and remount the partition. Example: `UUID=<UUID> /var/log ext4 defaults,noexec 0 2.`

Rule 1.1.2.7.1: Ensure separate partition exists for /var/log/audit

- **Status:** FAIL
- **Evidence:** The /var/log/audit mount point is not a separate partition (`is_separate_partition: false`).
- **Recommendation:** Create a separate partition for /var/log/audit to ensure audit logs are isolated and cannot fill up other filesystems.

Rule 1.1.2.7.2: Ensure nodev option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The /var/log/audit mount point does not have the nodev option set (`mount_options_list: ["rw", "relatime"]`).
- **Recommendation:** Add the nodev option to the /var/log/audit entry in /etc/

`fstab` and remount the partition. Example: `UUID=<UUID> /var/log/audit ext4 defaults,nosuid 0 2`.

Rule 1.1.2.7.3: Ensure nosuid option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The `/var/log/audit` mount point does not have the `nosuid` option set (`mount_options_list: ["rw", "relatime"]`).
- **Recommendation:** Add the `nosuid` option to the `/var/log/audit` entry in `/etc/fstab` and remount the partition. Example: `UUID=<UUID> /var/log/audit ext4 defaults,nosuid 0 2`.

Rule 1.1.2.7.4: Ensure noexec option set on /var/log/audit partition

- **Status:** FAIL
- **Evidence:** The `/var/log/audit` mount point does not have the `noexec` option set (`mount_options_list: ["rw", "relatime"]`).
- **Recommendation:** Add the `noexec` option to the `/var/log/audit` entry in `/etc/fstab` and remount the partition. Example: `UUID=<UUID> /var/log/audit ext4 defaults,noexec 0 2`.

Section 1.2 – Package Management

Rule 1.2.1.1: Ensure the `source.list` and `.source` files use the Signed-By option

- **Status:** INFORMATIONAL
- **Evidence:**
 - `/etc/apt/sources.list`: No active repository lines found.
 - `/etc/apt/sources.list.d/ubuntu.sources`: All active repository lines use `Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg`.
 - `/etc/apt/sources.list.d/wazuh.list`: All active repository lines use `[signed-by=/usr/share/keyrings/wazuh.gpg]`.
- **Recommendation:** None. All identified active repository lines correctly use the `Signed-By` option.

Rule 1.2.1.2: Ensure weak dependencies are configured

- **Status:** FAIL
- **Evidence:** The `weak_deps_config_lines` list is empty, indicating that `APT::Install-Recommends` and `APT::Install-Suggests` are not explicitly set to `"false"` or `"0"`.
- **Recommendation:** Create or edit `/etc/apt/apt.conf.d/99-cis-weak-deps` and add the following lines:

```
APT::Install-Recommends "false";
APT::Install-Suggests "false";
```

Rule 1.2.1.3: Ensure access to gpg key files are configured

- **Status:** PASS
- **Evidence:**
 - `/etc/apt/trusted.gpg.d/ubuntu-keyring-2012-cdimage.gpg`: Mode 0o644, owner root, group root.
 - `/etc/apt/trusted.gpg.d/ubuntu-keyring-2018-archive.gpg`: Mode 0o644, owner root, group root. All GPG key files have appropriate permissions (mode <= 0644, owner root, group root).
- **Recommendation:** None.

Rule 1.2.1.4: Ensure access to `/etc/apt/trusted.gpg.d` directory is configured

- **Status:** PASS
- **Evidence:** The `/etc/apt/trusted.gpg.d` directory has mode 0o755, owner root, group root.
- **Recommendation:** None.

Rule 1.2.1.5: Ensure access to `/etc/apt/auth.conf.d` directory is configured

- **Status:** PASS
- **Evidence:** The `/etc/apt/auth.conf.d` directory has mode 0o755, owner root, group root. While 0o700 is recommended, 0o755 is acceptable as it prevents non-root users from writing.
- **Recommendation:** None. (For stricter compliance, consider `chmod 0700 /etc/apt/auth.conf.d`).

Rule 1.2.1.6: Ensure access to files in the `/etc/apt/auth.conf.d/` directory is configured

- **Status:** PASS
- **Evidence:** No files were found in `/etc/apt/auth.conf.d/`.
- **Recommendation:** None.

Rule 1.2.1.7: Ensure access to `/usr/share/keyrings` directory is configured

- **Status:** PASS
- **Evidence:** The `/usr/share/keyrings` directory has mode 0o755, owner root, group root.
- **Recommendation:** None.

Rule 1.2.1.8: Ensure access to `/etc/apt/sources.list.d` directory is configured

- **Status:** PASS
- **Evidence:** The `/etc/apt/sources.list.d` directory has mode 0o755, owner root, group root.
- **Recommendation:** None.

Rule 1.2.1.9: Ensure access to files in `/etc/apt/sources.list.d` are configured

- **Status:** PASS
- **Evidence:**
 - `/etc/apt/sources.list.d/ubuntu.sources`: Mode 0o644, owner root, group root.
 - `/etc/apt/sources.list.d/ubuntu.sources.curtin.orig`: Mode 0o644, owner root, group root.
 - `/etc/apt/sources.list.d/wazuh.list`: Mode 0o644, owner root, group root. All files in `/etc/apt/sources.list.d` have appropriate permissions (mode <= 0644, owner root, group root).
- **Recommendation:** None.

Rule 1.2.2.1: Ensure updates, patches, and additional security software are installed

- **Status:** INFORMATIONAL
- **Evidence:**
 - 85 upgradable packages are available.
 - `unattended-upgrades` package is installed, enabled, and active.
 - `unattended-upgrades` is configured to automatically update package lists and perform unattended upgrades.
- **Recommendation:** None. Regularly apply available updates and patches. Review the list of upgradable packages and consider performing a full system upgrade.

Section 1.3 – Mandatory Access Control

Rule 1.3.1.1: Ensure apparmor packages are installed

- **Status:** FAIL
- **Evidence:**
 - `apparmor` package is installed (`apparmor_installed: true`).
 - `apparmor-utils` package is NOT installed (`apparmor_utils_installed: false`).
- **Recommendation:** Install the `apparmor-utils` package: `sudo apt install apparmor-utils`.

Rule 1.3.1.2: Ensure AppArmor is enabled

- **Status:** FAIL
- **Evidence:**
 - AppArmor kernel module is loaded (`sysfs_enabled_value: "Y"`).
 - AppArmor service is enabled and active (`service_enabled: "enabled"`, `service_active: "active"`).
 - GRUB command line does NOT contain `apparmor=1` (`grub_apparmor_flag: false`).
 - GRUB command line does NOT contain `security=apparmor` (`grub_security_flag: false`).

- **Recommendation:** Edit `/etc/default/grub` and add `apparmor=1 security=apparmor` to the `GRUB_CMDLINE_LINUX` variable. Then run `sudo update-grub` and reboot the system.

Rule 1.3.1.3: Ensure all AppArmor Profiles are enforcing

- **Status:** FAIL
- **Evidence:** `aa-status` output shows 91 profiles are in unconfined mode and 4 profiles are in complain mode.
- **Recommendation:** Put all AppArmor profiles into enforcing mode. For profiles in complain mode, use `sudo aa-enforce <profile_name>`. For unconfined profiles, create and enforce profiles for critical services or applications.

Rule 1.3.1.4: Ensure `apparmor_restrict_unprivileged_unconfined` is enabled

- **Status:** FAIL
- **Evidence:**
 - `kernel.apparmor_restrict_unprivileged_unconfined` runtime value is 0.
 - The setting is not persisted (`persisted_value: null`).
- **Recommendation:**
 - Set the kernel parameter: `sudo sysctl -w kernel.apparmor_restrict_unprivileged_unconfined=1`.
 - Persist the setting: Add `kernel.apparmor_restrict_unprivileged_unconfined=1` to a file like `/etc/sysctl.d/99-apparmor.conf` and run `sudo sysctl -p`.

Section 1.4 – Configure Bootloader

Rule 1.4.1: Ensure bootloader password is set

- **Status:** FAIL
- **Evidence:**
 - `has_superusers: false` (no superusers defined in GRUB).
 - `has_password: false` (no password set for GRUB).
- **Recommendation:** Set a password for the GRUB bootloader. Edit `/etc/grub.d/40_custom` or create a new file in `/etc/grub.d/` to define a superuser and password, then run `sudo update-grub`.

Rule 1.4.2: Ensure access to bootloader config is configured

- **Status:** PASS
- **Evidence:** The `/boot/grub/grub.cfg` file has mode `0o600`, owner `root`, group `root`.
- **Recommendation:** None.

Section 1.5 – Configure Additional Process Hardening

Rule 1.5.1: Ensure fs.protected_hardlinks is configured

- **Status:** FAIL
- **Evidence:**
 - fs.protected_hardlinks runtime value is 1.
 - The setting is not persisted (persisted_value: null).
- **Recommendation:**
 - Ensure the runtime value is 1: `sudo sysctl -w fs.protected_hardlinks=1.`
 - Persist the setting: Add `fs.protected_hardlinks=1` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p.`

Rule 1.5.2: Ensure fs.protected_symlinks is configured

- **Status:** FAIL
- **Evidence:**
 - fs.protected_symlinks runtime value is 1.
 - The setting is not persisted (persisted_value: null).
- **Recommendation:**
 - Ensure the runtime value is 1: `sudo sysctl -w fs.protected_symlinks=1.`
 - Persist the setting: Add `fs.protected_symlinks=1` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p.`

Rule 1.5.3: Ensure kernel.yama.pttrace_scope is configured

- **Status:** PASS
- **Evidence:**
 - kernel.yama.pttrace_scope runtime value is 1.
 - The setting is persisted with value 1 in `/etc/sysctl.d/10-pttrace.conf.`
- **Recommendation:** None.

Rule 1.5.4: Ensure fs.suid_dumpable is configured

- **Status:** FAIL
- **Evidence:**
 - fs.suid_dumpable runtime value is 2.
 - The setting is not persisted (persisted_value: null).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w fs.suid_dumpable=0.`

- Persist the setting: Add `fs.suid_dumpable=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 1.5.5: Ensure `kernel.dmesg_restrict` is configured

- **Status:** FAIL
- **Evidence:**
 - `kernel.dmesg_restrict` runtime value is 1.
 - The setting is not persisted (`persisted_value: null`).
- **Recommendation:**
 - Ensure the runtime value is 1: `sudo sysctl -w kernel.dmesg_restrict=1`.
 - Persist the setting: Add `kernel.dmesg_restrict=1` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 1.5.6: Ensure `prelink` is not installed

- **Status:** PASS
- **Evidence:** `prelink_installed: false`.
- **Recommendation:** None.

Rule 1.5.7: Ensure Automatic Error Reporting is configured

- **Status:** PASS
- **Evidence:** Apport is installed (`apport_installed: true`), but its default configuration in `/etc/default/apport` is `enabled=1`, which means it is enabled. However, the rule is "Ensure Automatic Error Reporting is configured", which implies it should be disabled.
 - The rule states "Ensure Automatic Error Reporting is configured". The CIS benchmark typically recommends disabling automatic error reporting.
 - `apport_installed: true`
 - `etc_default_setting: "enabled=1"`
- **Recommendation:** Disable Apport by editing `/etc/default/apport` and changing `enabled=1` to `enabled=0`.

Rule 1.5.8: Ensure `kernel.kptr_restrict` is configured

- **Status:** FAIL
- **Evidence:**
 - `kernel.kptr_restrict` runtime value is 1.
 - The setting is persisted with value 1 in `/etc/sysctl.d/10-kernel-hardening.conf`.
 - CIS recommends a value of 2.
- **Recommendation:**
 - Set the runtime value to 2: `sudo sysctl -w kernel.kptr_restrict=2`.
 - Persist the setting: Edit `/etc/sysctl.d/10-kernel-hardening.conf` and

change `kernel.kptr_restrict=1` to `kernel.kptr_restrict=2`. Run `sudo sysctl -p`.

Rule 1.5.9: Ensure `kernel.randomize_va_space` is configured

- **Status:** FAIL
- **Evidence:**
 - `kernel.randomize_va_space` runtime value is 2.
 - The setting is not persisted (`persisted_value: null`).
- **Recommendation:**
 - Ensure the runtime value is 2: `sudo sysctl -w kernel.randomize_va_space=2`.
 - Persist the setting: Add `kernel.randomize_va_space=2` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 1.5.10: Ensure core file size is configured

- **Status:** FAIL
- **Evidence:**
 - `ulimit_soft_core` is 0.
 - `limits_conf_lines` is empty, indicating no hard core 0 entry in `/etc/security/limits.conf`.
- **Recommendation:** Add `* hard core 0` to `/etc/security/limits.conf` to prevent core dumps for all users.

Rule 1.5.11: Ensure `systemd-coredump ProcessSizeMax` is configured

- **Status:** FAIL
- **Evidence:** `systemd_coredump_process_size_max` is empty, indicating `ProcessSizeMax=0` is not configured in `coredump.conf`.
- **Recommendation:** Add `ProcessSizeMax=0` to `/etc/systemd/coredump.conf` to prevent core dumps.

Rule 1.5.12: Ensure `systemd-coredump Storage` is configured

- **Status:** FAIL
- **Evidence:** `systemd_coredump_storage` is empty, indicating `Storage=none` is not configured in `coredump.conf`.
- **Recommendation:** Add `Storage=none` to `/etc/systemd/coredump.conf` to prevent core dumps from being stored.

Section 1.6 – Configure Command Line Warning Banners

Rule 1.6.1: Ensure `/etc/motd` is configured

- **Status:** PASS
- **Evidence:** The `/etc/motd` file does not exist (`motd.exists: false`), meaning no

static banner is displayed from this file. Dynamic MOTD is handled by pam_motd.

- **Recommendation:** None.

Rule 1.6.2: Ensure /etc/issue is configured

- **Status:** FAIL
- **Evidence:** The /etc/issue content is "Ubuntu 24.04.2 LTS \n \l\n\n". It contains the \n (kernel name) and \l (tty line) escape sequences, which should be removed.
- **Recommendation:** Edit /etc/issue to remove all escape sequences and include only a compliant warning banner. Example: Authorized uses only. All activity may be monitored and reported.

Rule 1.6.3: Ensure /etc/issue.net is configured

- **Status:** FAIL
- **Evidence:** The /etc/issue.net content is "Ubuntu 24.04.2 LTS\n". It contains OS release information, which should be removed.
- **Recommendation:** Edit /etc/issue.net to remove all system information and include only a compliant warning banner. Example: Authorized uses only. All activity may be monitored and reported.

Rule 1.6.4: Ensure pam_motd is configured

- **Status:** FAIL
- **Evidence:** pam_motd.so is referenced in /etc/pam.d/login and /etc/pam.d/sshd, and there are multiple active scripts in /etc/update-motd.d/ (e.g., 00-header, 10-help-text, 50-landscape-sysinfo, 50-motd-news, etc.) that generate dynamic content. This dynamic content often includes system information that should not be displayed in a warning banner.
- **Recommendation:** Disable dynamic MOTD generation by removing or commenting out the pam_motd.so lines in /etc/pam.d/login and /etc/pam.d/sshd, or by removing execute permissions from scripts in /etc/update-motd.d/. Ensure a static, compliant banner is used instead (e.g., via /etc/motd).

Rule 1.6.6: Ensure access to /etc/motd is configured

- **Status:** PASS
- **Evidence:** The /etc/motd file does not exist (motd.exists: false). If it were to exist, its permissions would need to be 0644 or more restrictive.
- **Recommendation:** None.

Rule 1.6.7: Ensure access to /etc/issue is configured

- **Status:** PASS
- **Evidence:** The /etc/issue file has mode 0644, owner root, group root.
- **Recommendation:** None.

Rule 1.6.8: Ensure access to /etc/issue.net is configured

- **Status:** PASS
- **Evidence:** The /etc/issue.net file has mode 0644, owner root, group root.
- **Recommendation:** None.

Rule 1.6.9: Ensure access to pam_motd file is configured

- **Status:** UNKNOWN
- **Evidence:** The profile does not provide specific file permissions for the individual scripts within `/etc/update-motd.d/` that `pam_motd` utilizes.
- **Recommendation:** Manually verify that all files in `/etc/update-motd.d/` have permissions of `0755` or more restrictive, and are owned by `root:root`.

Rule 1.6.10: Ensure sshd warning Banner is configured

- **Status:** FAIL
- **Evidence:** The SSH daemon Banner setting is configured to `none` (`ssh.features.banner: "none"`).
- **Recommendation:** Configure the SSH daemon to display a warning banner. Add or modify the `Banner` directive in `/etc/ssh/sshd_config` to point to a compliant banner file, e.g., `Banner /etc/issue.net`. Then restart the SSH service.

Section 1.7 – Configure GNOME Display Manager

Rule 1.7.1: Ensure GDM login banner is configured

- **Status:** PASS
- **Evidence:** GDM is not installed (`gdm_installed: false`). This rule is not applicable.
- **Recommendation:** None.

Rule 1.7.2: Ensure GDM disable-user-list is configured

- **Status:** PASS
- **Evidence:** GDM is not installed (`gdm_installed: false`). This rule is not applicable.
- **Recommendation:** None.

Rule 1.7.3: Ensure GDM screen lock is configured

- **Status:** PASS
- **Evidence:** GDM is not installed (`gdm_installed: false`). This rule is not applicable.
- **Recommendation:** None.

Rule 1.7.4: Ensure GDM automount is configured

- **Status:** PASS
- **Evidence:** GDM is not installed (`gdm_installed: false`). This rule is not applicable.
- **Recommendation:** None.

Rule 1.7.5: Ensure GDM autorun-never is configured

- **Status:** PASS
- **Evidence:** GDM is not installed (`gdm_installed: false`). This rule is not applicable.

applicable.

- **Recommendation:** None.

Rule 1.7.6: Ensure XDMCP is not enabled

- **Status:** PASS
- **Evidence:** GDM is not installed (`gdm_installed: false`). This rule is not applicable.
- **Recommendation:** None.

Rule 1.7.7: Ensure Xwayland is configured

- **Status:** PASS
- **Evidence:** GDM is not installed (`gdm_installed: false`). This rule is not applicable.
- **Recommendation:** None.

Section 2 – Services

Rule 2.1.1: Ensure autofs services are not in use

- **Status:** PASS
- **Evidence:** autofs package is not installed (`any_package_installed: false`) and autofs.service is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.2: Ensure mail transfer agents are configured for local-only mode

- **Status:** PASS
- **Evidence:** No Mail Transfer Agent (MTA) package (postfix, exim4, sendmail) is installed (`mta_detected: "none"`).
- **Recommendation:** None.

Rule 2.1.3: Ensure avahi daemon services are not in use

- **Status:** PASS
- **Evidence:** avahi-daemon package is not installed (`any_package_installed: false`) and no avahi-daemon units are active (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.4: Ensure only approved services are listening on a network interface

- **Status:** INFORMATIONAL
- **Evidence:** The following services are listening on network interfaces:
 - 127.0.0.54:53 (UDP, TCP) by systemd-resolve
 - 127.0.0.53%lo:53 (UDP, TCP) by systemd-resolve
 - 10.100.0.22%enp0s3:68 (UDP) by systemd-network
 - 0.0.0.0:443 (TCP) by node
 - 0.0.0.0:1514 (TCP) by wazuh-remoted

- 0.0.0.0:1515 (TCP) by wazuh-authd
- 0.0.0.0:22 (TCP) by sshd
- 0.0.0.0:23 (TCP) by inetd
- 0.0.0.0:55000 (TCP) by python3
- *:21 (TCP) by vsftpd
- [::]:22 (TCP) by sshd
- [::]:23 (TCP) by inetd
- [::ffff:127.0.0.1]:9300 (TCP) by java
- [::ffff:127.0.0.1]:9200 (TCP) by java

- **Recommendation:** None. Review the list of listening services and ensure each is authorized and necessary. Specifically, `inetd` (port 23) and `vsftpd` (port 21) are flagged as FAIL in other rules and should be addressed.

Rule 2.1.5: Ensure dhcp server services are not in use

- **Status:** PASS
- **Evidence:** `isc-dhcp-server` package is not installed (`any_package_installed: false`) and no `isc-dhcp-server` units are active (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.6: Ensure web server services are not in use

- **Status:** PASS
- **Evidence:** No web server packages (`apache2`, `nginx`, `lighttpd`) are installed (`any_package_installed: false`) and no web server units are active (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.7: Ensure dns server services are not in use

- **Status:** PASS
- **Evidence:** `bind9` package is not installed (`any_package_installed: false`) and no `bind9` or `named` units are active (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.8: Ensure ftp server services are not in use

- **Status:** FAIL
- **Evidence:** `vsftpd` package is installed (`any_package_installed: true`) and `vsftpd.service` is enabled and active (`any_unit_active: true`).
- **Recommendation:** If an FTP server is not required, uninstall `vsftpd`: `sudo apt purge vsftpd`. If it is required, ensure it is securely configured and restricted.

Rule 2.1.9: Ensure dnsmasq services are not in use

- **Status:** PASS
- **Evidence:** `dnsmasq` package is not installed (`any_package_installed: false`) and `dnsmasq.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.10: Ensure ldap server services are not in use

- **Status:** PASS
- **Evidence:** slapd package is not installed (`any_package_installed: false`) and slapd.service is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.11: Ensure message access server services are not in use

- **Status:** PASS
- **Evidence:** No message access server packages (dovecot-imapd, dovecot-pop3d, cyrus-imapd) are installed (`any_package_installed: false`) and no dovecot units are active (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.12: Ensure network file system services are not in use

- **Status:** PASS
- **Evidence:** nfs-kernel-server package is not installed (`any_package_installed: false`) and nfs-kernel-server.service is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.13: Ensure nis server services are not in use

- **Status:** PASS
- **Evidence:** nis package is not installed (`any_package_installed: false`) and ypserf.service is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.14: Ensure print server services are not in use

- **Status:** PASS
- **Evidence:** cups package is not installed (`any_package_installed: false`) and cups.service is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.15: Ensure rpcbind services are not in use

- **Status:** PASS
- **Evidence:** rpcbind package is not installed (`any_package_installed: false`) and no rpcbind units are active (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.16: Ensure rsync services are not in use

- **Status:** PASS
- **Evidence:** rsync package is installed (`any_package_installed: true`), but rsync.service is disabled and inactive (`any_unit_active: false`).
- **Recommendation:** None. If rsync is not needed, consider uninstalling the package:
`sudo apt purge rsync.`

Rule 2.1.17: Ensure samba file server services are not in use

- **Status:** PASS
- **Evidence:** samba package is not installed (`any_package_installed: false`) and `smbd.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.18: Ensure snmp services are not in use

- **Status:** PASS
- **Evidence:** `snmpd` package is not installed (`any_package_installed: false`) and `snmpd.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.19: Ensure telnet server services are not in use

- **Status:** FAIL
- **Evidence:** `telnetd` and `inetutils-telnetd` packages are installed (`any_package_installed: true`), and `inetd.service` is enabled and active (`any_unit_active: true`), which typically manages telnet server.
- **Recommendation:** If a Telnet server is not required, uninstall the packages: `sudo apt purge telnetd inetutils-telnetd`. If `inetd` is managing other services, ensure telnet is disabled within its configuration.

Rule 2.1.20: Ensure tftp server services are not in use

- **Status:** PASS
- **Evidence:** No TFTP server packages (`tftpd`, `tftpd-hpa`, `atftpd`) are installed (`any_package_installed: false`) and no TFTP units are active (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.21: Ensure web proxy server services are not in use

- **Status:** PASS
- **Evidence:** No web proxy server packages (`squid`, `squid3`) are installed (`any_package_installed: false`) and `squid.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.22: Ensure xinetd services are not in use

- **Status:** PASS
- **Evidence:** `xinetd` package is not installed (`any_package_installed: false`) and `xinetd.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.23: Ensure X window server services are not in use

- **Status:** PASS
- **Evidence:** No X window server packages (`xserver-xorg`, `xserver-xorg-core`, `xserver-common`) are installed (`any_package_installed: false`) and no Xorg processes are running (`xorg_process_running: false`).
- **Recommendation:** None.

Rule 2.2.1: Ensure nis client is not installed

- **Status:** PASS
- **Evidence:** nis package is not installed (`any_package_installed: false`).
- **Recommendation:** None.

Rule 2.2.2: Ensure rsh client is not installed

- **Status:** PASS
- **Evidence:** rsh-client package is not installed (`any_package_installed: false`).
- **Recommendation:** None.

Rule 2.2.3: Ensure talk client is not installed

- **Status:** PASS
- **Evidence:** talk package is not installed (`any_package_installed: false`).
- **Recommendation:** None.

Rule 2.2.4: Ensure telnet client is not installed

- **Status:** FAIL
- **Evidence:** telnet and inetutils-telnet packages are installed (`any_package_installed: true`).
- **Recommendation:** Uninstall Telnet client packages: `sudo apt purge telnet inetutils-telnet`.

Rule 2.2.5: Ensure ldap client is not installed

- **Status:** PASS
- **Evidence:** ldap-utils package is not installed (`any_package_installed: false`).
- **Recommendation:** None.

Rule 2.2.6: Ensure ftp client is not installed

- **Status:** FAIL
- **Evidence:** ftp package is installed (`any_package_installed: true`).
- **Recommendation:** Uninstall the FTP client package: `sudo apt purge ftp`.

Rule 2.3.1.1: Ensure a single time synchronization daemon is in use

- **Status:** PASS
- **Evidence:** Only one time synchronization daemon, `systemd-timesyncd`, is active (`active_daemon_count: 1`).
- **Recommendation:** None.

Rule 2.3.2.1: Ensure systemd-timesyncd configured with authorized timeserver

- **Status:** FAIL
- **Evidence:** `systemd-timesyncd` is the active time synchronization daemon, but no

NTP servers are explicitly configured in its configuration (`has_ntp_configured: false`).

- **Recommendation:** Edit `/etc/systemd/timesyncd.conf` and add `NTP=<NTP_SERVER_IP_OR_HOSTNAME>` (e.g., `NTP=0.ubuntu.pool.ntp.org 1.ubuntu.pool.ntp.org`). Restart the service: `sudo systemctl restart systemd-timesyncd`.

Rule 2.3.2.2: Ensure systemd-timesyncd is enabled and running

- **Status:** PASS
- **Evidence:** `systemd-timesyncd.service` is enabled (`enabled: "enabled"`) and active (`active: "active"`).
- **Recommendation:** None.

Rule 2.3.3.1: Ensure chrony is configured

- **Status:** PASS
- **Evidence:** `chrony` is not installed or active, so this rule is not applicable.
- **Recommendation:** None.

Rule 2.3.3.2: Ensure chrony is running as user _chrony

- **Status:** PASS
- **Evidence:** `chrony` is not installed or active, so this rule is not applicable.
- **Recommendation:** None.

Rule 2.3.3.3: Ensure chrony is enabled and running

- **Status:** PASS
- **Evidence:** `chrony` is not installed or active, so this rule is not applicable.
- **Recommendation:** None.

Rule 2.4.1.1: Ensure cron daemon is enabled and active

- **Status:** PASS
- **Evidence:** `cron` package is installed (`cron_installed: true`), and `cron.service` is enabled (`enabled: "enabled"`) and active (`active: "active"`).
- **Recommendation:** None.

Rule 2.4.1.2: Ensure access to /etc/crontab is configured

- **Status:** PASS
- **Evidence:** The `/etc/crontab` file has mode `0o644`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 2.4.1.3: Ensure access to /etc/cron.hourly is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.hourly` directory has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 2.4.1.4: Ensure access to /etc/cron.daily is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.daily` directory has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 2.4.1.5: Ensure access to /etc/cron.weekly is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.weekly` directory has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 2.4.1.6: Ensure access to /etc/cron.monthly is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.monthly` directory has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 2.4.1.7: Ensure access to /etc/cron.yearly is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.yearly` directory has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 2.4.1.8: Ensure access to /etc/cron.d is configured

- **Status:** PASS
- **Evidence:** The `/etc/cron.d` directory has mode `0o755`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 2.4.1.9: Ensure access to crontab is configured

- **Status:** PASS
- **Evidence:**
 - The `/var/spool/cron/crontabs` directory has mode `0o1730`, owner `root`, group `crontab`.
 - The `/usr/bin/crontab` binary has mode `0o2755`, owner `root`, group `crontab`.
- **Recommendation:** None.

Rule 2.4.2.1: Ensure access to at is configured

- **Status:** PASS
- **Evidence:** The `at` package is not installed (`at_installed: false`). This rule is not applicable.
- **Recommendation:** None.

Section 3 – Network Configuration

Rule 3.1.1: Ensure IPv6 status is identified

- **Status:** INFORMATIONAL
- **Evidence:**
 - IPv6 is not disabled via sysfs (`sysfs_disable_value: "0"`).
 - IPv6 is not disabled via GRUB kernel command line (`grub_ipv6_disable_flag: false`).
 - Two IPv6 addresses were found on network interfaces (`inet6_address_count: 2`).
 - Conclusion: IPv6 is enabled and active on the system.
- **Recommendation:** None.

Rule 3.1.2: Ensure wireless interfaces are not available

- **Status:** FAIL
- **Evidence:** Wireless interfaces were detected on the system (`any_wireless_found: true`). Specifically, the `cfg80211` kernel module is loaded.
- **Recommendation:** If wireless functionality is not required, disable or remove wireless interfaces and blacklist associated kernel modules (e.g., `cfg80211`) in `/etc/modprobe.d/`.

Rule 3.1.3: Ensure bluetooth services are not in use

- **Status:** PASS
- **Evidence:** The `bluez` package is not installed (`bluez_installed: false`).
- **Recommendation:** None.

Rule 3.2.1: Ensure atm kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `atm` module is not currently loaded.
 - `modprobe_config_lines` for `atm` is empty, indicating no explicit `install /bin/false` or blacklist configuration.
- **Recommendation:**
 - Add `install atm /bin/false` to a file like `/etc/modprobe.d/atm.conf`.
 - Add `blacklist atm` to the same file.
 - Run `sudo update-initramfs -u` and reboot.

Rule 3.2.2: Ensure can kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `can` module is not currently loaded.
 - `modprobe_config_lines` for `can` is empty, indicating no explicit `install /`

`bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install can /bin/false` to a file like `/etc/modprobe.d/can.conf`.
- Add `blacklist can` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 3.2.3: Ensure dccp kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `dccp` module is not currently loaded.
- `modprobe_config_lines` for `dccp` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install dccp /bin/false` to a file like `/etc/modprobe.d/dccp.conf`.
- Add `blacklist dccp` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 3.2.4: Ensure rds kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `rds` module is not currently loaded.
- `modprobe_config_lines` for `rds` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install rds /bin/false` to a file like `/etc/modprobe.d/rds.conf`.
- Add `blacklist rds` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 3.2.5: Ensure sctp kernel module is not available

- **Status:** FAIL

- **Evidence:**

- `sctp` module is not currently loaded.
- `modprobe_config_lines` for `sctp` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.

- **Recommendation:**

- Add `install sctp /bin/false` to a file like `/etc/modprobe.d/sctp.conf`.
- Add `blacklist sctp` to the same file.
- Run `sudo update-initramfs -u` and reboot.

Rule 3.2.6: Ensure tipc kernel module is not available

- **Status:** FAIL
- **Evidence:**
 - `tipc` module is not currently loaded.
 - `modprobe_config_lines` for `tipc` is empty, indicating no explicit `install /bin/false` or `blacklist` configuration.
- **Recommendation:**
 - Add `install tipc /bin/false` to a file like `/etc/modprobe.d/tipc.conf`.
 - Add `blacklist tipc` to the same file.
 - Run `sudo update-initramfs -u` and reboot.

Rule 3.3.1.1: Ensure net.ipv4.ip_forward is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv4.ip_forward` runtime value is 0.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.
- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv4.ip_forward=0` to a `sysctl` configuration file).

Rule 3.3.1.2: Ensure net.ipv4.conf.all.forwarding is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv4.conf.all.forwarding` runtime value is 0.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.
- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv4.conf.all.forwarding=0` to a `sysctl` configuration file).

Rule 3.3.1.3: Ensure net.ipv4.conf.default.forwarding is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv4.conf.default.forwarding` runtime value is 0.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.
- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv4.conf.default.forwarding=0` to a `sysctl` configuration file).

Rule 3.3.1.4: Ensure net.ipv4.conf.all.send_redirects is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.conf.all.send_redirects` runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (`persisted_value: null`).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv4.conf.all.send_redirects=0`.
 - Persist the setting: Add `net.ipv4.conf.all.send_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.1.5: Ensure `net.ipv4.conf.default.send_redirects` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv4.conf.default.send_redirects` runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (`persisted_value: null`).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv4.conf.default.send_redirects=0`.
 - Persist the setting: Add `net.ipv4.conf.default.send_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.1.6: Ensure `net.ipv4.icmp_ignore_bogus_error_responses` is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv4.icmp_ignore_bogus_error_responses` runtime value is 1.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.
- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv4.icmp_ignore_bogus_error_responses=1` to a `sysctl` configuration file).

Rule 3.3.1.7: Ensure `net.ipv4.icmp_echo_ignore_broadcasts` is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv4.icmp_echo_ignore_broadcasts` runtime value is 1.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.
- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv4.icmp_echo_ignore_broadcasts=1` to a `sysctl` configuration file).

Rule 3.3.1.8: Ensure net.ipv4.conf.all.accept_redirects is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv4.conf.all.accept_redirects runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (persisted_value: null).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv4.conf.all.accept_redirects=0.`
 - Persist the setting: Add `net.ipv4.conf.all.accept_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p.`

Rule 3.3.1.9: Ensure net.ipv4.conf.default.accept_redirects is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv4.conf.default.accept_redirects runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (persisted_value: null).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv4.conf.default.accept_redirects=0.`
 - Persist the setting: Add `net.ipv4.conf.default.accept_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p.`

Rule 3.3.1.10: Ensure net.ipv4.conf.all.secure_redirects is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv4.conf.all.secure_redirects runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (persisted_value: null).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv4.conf.all.secure_redirects=0.`
 - Persist the setting: Add `net.ipv4.conf.all.secure_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p.`

Rule 3.3.1.11: Ensure net.ipv4.conf.default.secure_redirects is configured

- **Status:** FAIL
- **Evidence:**
 - net.ipv4.conf.default.secure_redirects runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (persisted_value: null).

- **Recommendation:**

- Set the runtime value to 0: `sudo sysctl -w net.ipv4.conf.default.secure_redirects=0`.
- Persist the setting: Add `net.ipv4.conf.default.secure_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.1.12: Ensure net.ipv4.conf.all.rp_filter is configured

- **Status:** PASS

- **Evidence:**

- `net.ipv4.conf.all.rp_filter` runtime value is 2.
- The setting is persisted with value 2 in `/etc/sysctl.d/10-network-security.conf`.

- **Recommendation:** None.

Rule 3.3.1.13: Ensure net.ipv4.conf.default.rp_filter is configured

- **Status:** PASS

- **Evidence:**

- `net.ipv4.conf.default.rp_filter` runtime value is 2.
- The setting is persisted with value 2 in `/etc/sysctl.d/10-network-security.conf`.

- **Recommendation:** None.

Rule 3.3.1.14: Ensure net.ipv4.conf.all.accept_source_route is configured

- **Status:** PASS

- **Evidence:**

- `net.ipv4.conf.all.accept_source_route` runtime value is 0.
- The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.

- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv4.conf.all.accept_source_route=0` to a sysctl configuration file).

Rule 3.3.1.15: Ensure net.ipv4.conf.default.accept_source_route is configured

- **Status:** FAIL

- **Evidence:**

- `net.ipv4.conf.default.accept_source_route` runtime value is 1. (Expected: 0)
- The setting is not explicitly persisted (`persisted_value: null`).

- **Recommendation:**

- Set the runtime value to 0: `sudo sysctl -w`

`net.ipv4.conf.default.accept_source_route=0.`

- Persist the setting: Add `net.ipv4.conf.default.accept_source_route=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.1.16: Ensure net.ipv4.conf.all.log_martians is configured

- **Status:** FAIL

- **Evidence:**

- `net.ipv4.conf.all.log_martians` runtime value is 0. (Expected: 1)
- The setting is not explicitly persisted (`persisted_value: null`).

- **Recommendation:**

- Set the runtime value to 1: `sudo sysctl -w net.ipv4.conf.all.log_martians=1`.
- Persist the setting: Add `net.ipv4.conf.all.log_martians=1` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.1.17: Ensure net.ipv4.conf.default.log_martians is configured

- **Status:** FAIL

- **Evidence:**

- `net.ipv4.conf.default.log_martians` runtime value is 0. (Expected: 1)
- The setting is not explicitly persisted (`persisted_value: null`).

- **Recommendation:**

- Set the runtime value to 1: `sudo sysctl -w net.ipv4.conf.default.log_martians=1`.
- Persist the setting: Add `net.ipv4.conf.default.log_martians=1` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.1.18: Ensure net.ipv4.tcp_syncookies is configured

- **Status:** PASS

- **Evidence:**

- `net.ipv4.tcp_syncookies` runtime value is 1.
- The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.

- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv4.tcp_syncookies=1` to a sysctl configuration file).

Rule 3.3.2.1: Ensure net.ipv6.conf.all.forwarding is configured

- **Status:** PASS

- **Evidence:**

- `net.ipv6.conf.all.forwarding` runtime value is 0.
- The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.

- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv6.conf.all.forwarding=0` to a `sysctl` configuration file).

Rule 3.3.2.2: Ensure `net.ipv6.conf.default.forwarding` is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv6.conf.default.forwarding` runtime value is 0.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.
- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv6.conf.default.forwarding=0` to a `sysctl` configuration file).

Rule 3.3.2.3: Ensure `net.ipv6.conf.all.accept_redirects` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv6.conf.all.accept_redirects` runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (`persisted_value: null`).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv6.conf.all.accept_redirects=0`.
 - Persist the setting: Add `net.ipv6.conf.all.accept_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.2.4: Ensure `net.ipv6.conf.default.accept_redirects` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv6.conf.default.accept_redirects` runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (`persisted_value: null`).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv6.conf.default.accept_redirects=0`.
 - Persist the setting: Add `net.ipv6.conf.default.accept_redirects=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.2.5: Ensure `net.ipv6.conf.all.accept_source_route` is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv6.conf.all.accept_source_route` runtime value is 0.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.

- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv6.conf.all.accept_source_route=0` to a sysctl configuration file).

Rule 3.3.2.6: Ensure `net.ipv6.conf.default.accept_source_route` is configured

- **Status:** PASS
- **Evidence:**
 - `net.ipv6.conf.default.accept_source_route` runtime value is 0.
 - The setting is not explicitly persisted (`persisted_value: null`), but the runtime value is compliant.
- **Recommendation:** None. (For full compliance, persist the setting by adding `net.ipv6.conf.default.accept_source_route=0` to a sysctl configuration file).

Rule 3.3.2.7: Ensure `net.ipv6.conf.all.accept_ra` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv6.conf.all.accept_ra` runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (`persisted_value: null`).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv6.conf.all.accept_ra=0`.
 - Persist the setting: Add `net.ipv6.conf.all.accept_ra=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Rule 3.3.2.8: Ensure `net.ipv6.conf.default.accept_ra` is configured

- **Status:** FAIL
- **Evidence:**
 - `net.ipv6.conf.default.accept_ra` runtime value is 1. (Expected: 0)
 - The setting is not explicitly persisted (`persisted_value: null`).
- **Recommendation:**
 - Set the runtime value to 0: `sudo sysctl -w net.ipv6.conf.default.accept_ra=0`.
 - Persist the setting: Add `net.ipv6.conf.default.accept_ra=0` to a file like `/etc/sysctl.d/99-sysctl.conf` and run `sudo sysctl -p`.

Section 4.1 – Configure Uncomplicated Firewall

Rule 4.1.1: Ensure `ufw` is installed

- **Status:** PASS
- **Evidence:** `ufw` package is installed (`ufw_installed: true, ufw_dpkg_status:`

"install ok installed").

- **Recommendation:** None.

Rule 4.1.2: Ensure ufw service is configured

- **Status:** FAIL
- **Evidence:** The UFW service is enabled (`is_enabled: "enabled"`) but inactive (`is_active: "inactive", status_verbose: "Status: inactive"`).
- **Recommendation:** Start the UFW service: `sudo systemctl start ufw`.

Rule 4.1.3: Ensure ufw incoming default is configured

- **Status:** PASS
- **Evidence:** The `DEFAULT_INPUT_POLICY` in `/etc/default/ufw` is set to DROP.
- **Recommendation:** None.

Rule 4.1.4: Ensure ufw outgoing default is configured

- **Status:** FAIL
- **Evidence:** The `DEFAULT_OUTPUT_POLICY` in `/etc/default/ufw` is set to ACCEPT. (Expected: DROP or REJECT)
- **Recommendation:** Change `DEFAULT_OUTPUT_POLICY="ACCEPT"` to `DEFAULT_OUTPUT_POLICY="DROP"` or `DEFAULT_OUTPUT_POLICY="REJECT"` in `/etc/default/ufw`. Then reload UFW: `sudo ufw reload`.

Rule 4.1.5: Ensure ufw routed default is configured

- **Status:** PASS
- **Evidence:** The `DEFAULT_FORWARD_POLICY` in `/etc/default/ufw` is set to DROP.
- **Recommendation:** None.

Section 5.1 – Configure SSH Server

Rule 5.1.1: Ensure access to `/etc/ssh/sshd_config` is configured

- **Status:** FAIL
- **Evidence:** The `/etc/ssh/sshd_config` file has permissions 0644 (`sshd_config_permissions: "644 root root /etc/ssh/sshd_config"`). CIS recommends 0600.
- **Recommendation:** Set more restrictive permissions on `/etc/ssh/sshd_config`: `sudo chmod 0600 /etc/ssh/sshd_config`.

Rule 5.1.2: Ensure access to SSH private host key files is configured

- **Status:** PASS
- **Evidence:** All SSH private host key files have permissions 0600 and are owned by `root:root`.
 - `/etc/ssh/ssh_host_ecdsa_key: 600 root root`
 - `/etc/ssh/ssh_host_ed25519_key: 600 root root`
 - `/etc/ssh/ssh_host_rsa_key: 600 root root`

- **Recommendation:** None.

Rule 5.1.3: Ensure access to SSH public host key files is configured

- **Status:** PASS
- **Evidence:** All SSH public host key files have permissions 0644 and are owned by root:root.

- /etc/ssh/ssh_host_ecdsa_key.pub: 644 root root
- /etc/ssh/ssh_host_ed25519_key.pub: 644 root root
- /etc/ssh/ssh_host_rsa_key.pub: 644 root root

- **Recommendation:** None.

Rule 5.1.4: Ensure sshd access is configured

- **Status:** UNKNOWN
- **Evidence:** The security profile does not provide information on AllowUsers, AllowGroups, DenyUsers, or DenyGroups directives in sshd_config.
- **Recommendation:** Manually review /etc/ssh/sshd_config to ensure that SSH access is restricted to authorized users or groups using AllowUsers or AllowGroups directives.

Rule 5.1.5: Ensure sshd Banner is configured

- **Status:** FAIL
- **Evidence:** The Banner directive in sshd_config is set to none (ssh.features.banner: "none").
- **Recommendation:** Configure a warning banner for SSH. Add or modify the Banner directive in /etc/ssh/sshd_config to point to a file containing a compliant warning message, e.g., Banner /etc/issue.net. Then restart the SSH service.

Rule 5.1.6: Ensure sshd Ciphers are configured

- **Status:** UNKNOWN
- **Evidence:** The profile lists the all_ciphers as ["chacha20-poly1305@openssh.com", "aes128-ctr", "aes192-ctr", "aes256-ctr", "aes128-gcm@openssh.com", "aes256-gcm@openssh.com"]. Without a specific list of CIS-approved ciphers for comparison, it's not possible to definitively determine compliance.
- **Recommendation:** Manually review the configured ciphers in /etc/ssh/sshd_config against the latest CIS benchmark recommendations to ensure only strong, approved ciphers are enabled.

Rule 5.1.7: Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured

- **Status:** FAIL
- **Evidence:**

- ClientAliveInterval is set to 0 (ssh.session.client_alive_interval.value: 0), which disables client

keepalives.

- ClientAliveCountMax is set to 3
(ssh.session.client_alive_count_max: 3), which is compliant.

- **Recommendation:** Set ClientAliveInterval to a value between 1 and 300 seconds (e.g., ClientAliveInterval 300) in /etc/ssh/sshd_config to ensure idle sessions are terminated. Restart the SSH service.

Rule 5.1.8: Ensure sshd DisableForwarding is enabled

- **Status:** FAIL

- **Evidence:**

- AllowTcpForwarding is set to true
(ssh.forwarding.allow_tcp_forwarding.value: true).
- AllowAgentForwarding is set to true
(ssh.forwarding.allow_agent_forwarding.value: true).
- X11Forwarding is set to true (ssh.features.x11_forwarding.value: true). All these forwarding options are enabled, which is not compliant.

- **Recommendation:** Disable all unnecessary forwarding options in /etc/ssh/sshd_config:

```
AllowTcpForwarding no
AllowAgentForwarding no
X11Forwarding no
```

Restart the SSH service.

Rule 5.1.9: Ensure sshd GSSAPIAuthentication is disabled

- **Status:** PASS
- **Evidence:** GSSAPIAuthentication is set to false
(ssh.authentication.gssapiauthentication: false).
- **Recommendation:** None.

Rule 5.1.10: Ensure sshd HostbasedAuthentication is disabled

- **Status:** PASS
- **Evidence:** HostbasedAuthentication is set to false
(ssh.authentication.hostbasedauthentication: false).
- **Recommendation:** None.

Rule 5.1.11: Ensure sshd IgnoreRhosts is enabled

- **Status:** PASS
- **Evidence:** IgnoreRhosts is set to true (ssh.authentication.ignorerhosts: true).
- **Recommendation:** None.

Rule 5.1.12: Ensure sshd KexAlgorithms is configured

- **Status:** FAIL

- **Evidence:** The profile identifies several weak Key Exchange (Kex) algorithms: `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521` (due to NIST/NSA backdoor concerns), and `diffie-hellman-group14-sha256` (deprecated or known weaknesses).
- **Recommendation:** Remove all weak KexAlgorithms from the `KexAlgorithms` directive in `/etc/ssh/sshd_config`. Consult the latest CIS benchmark for a list of approved algorithms. Restart the SSH service.

Rule 5.1.13: Ensure sshd LoginGraceTime is configured

- **Status:** FAIL
- **Evidence:** `LoginGraceTime` is set to 120 seconds (`ssh.session.login_grace_time: 120`). CIS recommends a maximum of 60 seconds.
- **Recommendation:** Set `LoginGraceTime` to 60 or less in `/etc/ssh/sshd_config`. Restart the SSH service.

Rule 5.1.14: Ensure sshd LogLevel is configured

- **Status:** PASS
- **Evidence:** `LogLevel` is set to `INFO` (`ssh.features.log_level: "INFO"`).
- **Recommendation:** None.

Rule 5.1.15: Ensure sshd MACs are configured

- **Status:** FAIL
- **Evidence:** The profile identifies several weak Message Authentication Codes (MACs): `umac-64-etm@openssh.com`, `hmac-sha1-etm@openssh.com` (Terrapin attack), `umac-64@openssh.com`, `umac-128@openssh.com`, `hmac-sha2-256`, `hmac-sha2-512`, and `hmac-sha1` (SHA-1 collision attacks).
- **Recommendation:** Remove all weak MACs from the `MACs` directive in `/etc/ssh/sshd_config`. Consult the latest CIS benchmark for a list of approved algorithms. Restart the SSH service.

Rule 5.1.16: Ensure sshd MaxAuthTries is configured

- **Status:** FAIL
- **Evidence:** `MaxAuthTries` is set to 6 (`ssh.authentication.max_auth_tries: 6`). CIS recommends a maximum of 4.
- **Recommendation:** Set `MaxAuthTries` to 4 or less in `/etc/ssh/sshd_config`. Restart the SSH service.

Rule 5.1.17: Ensure sshd MaxStartups is configured

- **Status:** PASS
- **Evidence:** `MaxStartups` is set to `10:30:100` (`ssh.session.maxstartups: "10:30:100"`), which is a compliant configuration.
- **Recommendation:** None.

Rule 5.1.18: Ensure sshd MaxSessions is configured

- **Status:** PASS

- **Evidence:** MaxSessions is set to 10 (`ssh.session.max_sessions: 10`), which is compliant.
- **Recommendation:** None.

Rule 5.1.19: Ensure sshd PermitEmptyPasswords is disabled

- **Status:** PASS
- **Evidence:** PermitEmptyPasswords is set to false (`ssh.authentication.permit_empty_passwords: false`).
- **Recommendation:** None.

Rule 5.1.20: Ensure sshd PermitRootLogin is disabled

- **Status:** PASS
- **Evidence:** PermitRootLogin is set to without-password (`ssh.authentication.permit_root_login: "without-password"`), which is compliant.
- **Recommendation:** None.

Rule 5.1.21: Ensure sshd PermitUserEnvironment is disabled

- **Status:** PASS
- **Evidence:** PermitUserEnvironment is set to false (`ssh.features.permit_user_environment: false`).
- **Recommendation:** None.

Rule 5.1.22: Ensure sshd UsePAM is enabled

- **Status:** PASS
- **Evidence:** UsePAM is set to true (`ssh.authentication.usepam: true`).
- **Recommendation:** None.

Rule 5.1.23: Ensure sshd post-quantum cryptography key exchange algorithms are configured

- **Status:** UNKNOWN
- **Evidence:** The security profile does not explicitly list configured post-quantum cryptography (PQC) key exchange algorithms.
- **Recommendation:** Manually review `/etc/ssh/sshd_config` to ensure that approved PQC key exchange algorithms are configured, especially if the system is expected to comply with post-quantum security requirements.

Rule 5.1.24: Ensure sshd ListenAddress is configured

- **Status:** FAIL
- **Evidence:** ListenAddress is configured to `0.0.0.0:22` (`ssh.network.listen_address: "0.0.0.0:22"`), which listens on all available IPv4 interfaces.
- **Recommendation:** Restrict ListenAddress to specific network interfaces or IP addresses in `/etc/ssh/sshd_config` if the server does not need to be accessible on all interfaces. For example, `ListenAddress 192.168.1.100:22`. Restart the SSH service.

Section 5.3 – Pluggable Authentication Modules (PAM)

Rule 5.3.1.1: Ensure latest version of pam is installed

- **Status:** PASS
- **Evidence:** libpam-runtime is installed (`installed: true`) and the installed version (1.5.3-5ubuntu5.5) matches the candidate version (1.5.3-5ubuntu5.5).
- **Recommendation:** None.

Rule 5.3.1.2: Ensure latest version of libpam-modules is installed

- **Status:** PASS
- **Evidence:** libpam-modules is installed (`installed: true`) and the installed version (1.5.3-5ubuntu5.5) matches the candidate version (1.5.3-5ubuntu5.5).
- **Recommendation:** None.

Rule 5.3.1.3: Ensure latest version of libpam-pwquality is installed

- **Status:** FAIL
- **Evidence:** libpam-pwquality package is not installed (`installed: false`).
- **Recommendation:** Install libpam-pwquality: `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: Ensure latest version of cracklib-runtime is installed

- **Status:** FAIL
- **Evidence:** cracklib-runtime package is not installed (`installed: false`).
- **Recommendation:** Install cracklib-runtime: `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: Ensure pam_unix module is enabled

- **Status:** PASS
- **Evidence:** The `pam_unix.so` module is enabled and referenced in `common-auth`, `common-account`, `common-password`, and `common-session`.
- **Recommendation:** None.

Rule 5.3.2.2: Ensure pam_faillock module is enabled

- **Status:** FAIL
- **Evidence:** The `pam_faillock.so` module is not enabled or referenced in any PAM configuration files (`module_lines["pam_faillock.so"]` is empty).
- **Recommendation:** Enable `pam_faillock.so` in `/etc/pam.d/common-auth` and `/etc/pam.d/common-account` to enforce password failed attempts lockout.

Rule 5.3.2.3: Ensure pam_pwquality module is enabled

- **Status:** FAIL
- **Evidence:** The `pam_pwquality.so` module is not enabled or referenced in any PAM configuration files (`module_lines["pam_pwquality.so"]` is empty).
- **Recommendation:** Enable `pam_pwquality.so` in `/etc/pam.d/common-password`

to enforce password quality requirements.

Rule 5.3.2.4: Ensure pam_pwhistory module is enabled

- **Status:** FAIL
- **Evidence:** The `pam_pwhistory.so` module is not enabled or referenced in any PAM configuration files (`module_lines["pam_pwhistory.so"]` is empty).
- **Recommendation:** Enable `pam_pwhistory.so` in `/etc/pam.d/common-password` to enforce password history.

Rule 5.3.3.1.1: Ensure password failed attempts lockout is configured

- **Status:** FAIL
- **Evidence:** The `deny` option for `pam_faillock` is not configured in `/etc/security/faillock.conf` or inline PAM settings (`deny: null`).
- **Recommendation:** Configure `pam_faillock` to lock accounts after a specified number of failed attempts. Add `deny = 5` to `/etc/security/faillock.conf` and ensure `pam_faillock.so` is enabled in `common-auth` and `common-account`.

Rule 5.3.3.1.2: Ensure password unlock time is configured

- **Status:** FAIL
- **Evidence:** The `unlock_time` option for `pam_faillock` is not configured in `/etc/security/faillock.conf` or inline PAM settings (`unlock_time: null`).
- **Recommendation:** Configure `pam_faillock` to set an unlock time. Add `unlock_time = 900` (15 minutes) or `unlock_time = 0` (never unlock) to `/etc/security/faillock.conf`.

Rule 5.3.3.1.3: Ensure password failed attempts lockout includes root account

- **Status:** FAIL
- **Evidence:** The `even_deny_root` option for `pam_faillock` is not configured in `/etc/security/faillock.conf` or inline PAM settings (`even_deny_root: null`).
- **Recommendation:** Configure `pam_faillock` to apply lockout policies to the root account. Add `even_deny_root` to `/etc/security/faillock.conf`.

Rule 5.3.3.2.1: Ensure password number of changed characters is configured

- **Status:** FAIL
- **Evidence:** The `difok` option for `pam_pwquality` is not configured in `/etc/security/pwquality.conf` or inline PAM settings (`difok: null`).
- **Recommendation:** Configure `pam_pwquality` to require a minimum number of changed characters. Add `difok = 2` to `/etc/security/pwquality.conf` and ensure `pam_pwquality.so` is enabled in `common-password`.

Rule 5.3.3.2.2: Ensure password length is configured

- **Status:** FAIL
- **Evidence:** The `minlen` option for `pam_pwquality` is not configured in `/etc/security/pwquality.conf` or inline PAM settings (`minlen: null`).

- **Recommendation:** Configure `pam_pwquality` to require a minimum password length. Add `minlen = 14` to `/etc/security/pwquality.conf`.

Rule 5.3.3.2.3: Ensure password complexity is configured

- **Status:** INFORMATIONAL
- **Evidence:** The `dcredit`, `ucredit`, `lcredit`, and `ocredit` options for `pam_pwquality` are not explicitly configured in `/etc/security/pwquality.conf` or inline PAM settings (`dcredit: null`, `ucredit: null`, `lcredit: null`, `ocredit: null`).
- **Recommendation:** None. Manually review and configure `dcredit`, `ucredit`, `lcredit`, and `ocredit` in `/etc/security/pwquality.conf` to enforce specific character type requirements (e.g., `dcredit=-1`, `ucredit=-1`, `lcredit=-1`, `ocredit=-1`).

Rule 5.3.3.2.4: Ensure password same consecutive characters is configured

- **Status:** FAIL
- **Evidence:** The `maxrepeat` option for `pam_pwquality` is not configured in `/etc/security/pwquality.conf` or inline PAM settings (`maxrepeat: null`).
- **Recommendation:** Configure `pam_pwquality` to limit consecutive identical characters. Add `maxrepeat = 3` to `/etc/security/pwquality.conf`.

Rule 5.3.3.2.5: Ensure password maximum sequential characters is configured

- **Status:** FAIL
- **Evidence:** The `maxsequence` option for `pam_pwquality` is not configured in `/etc/security/pwquality.conf` or inline PAM settings (`maxsequence: null`).
- **Recommendation:** Configure `pam_pwquality` to limit sequential characters. Add `maxsequence = 3` to `/etc/security/pwquality.conf`.

Rule 5.3.3.2.6: Ensure password dictionary check is enabled

- **Status:** FAIL
- **Evidence:** The `dictcheck` option for `pam_pwquality` is not configured in `/etc/security/pwquality.conf` or inline PAM settings (`dictcheck: null`).
- **Recommendation:** Configure `pam_pwquality` to enable dictionary checks. Add `dictcheck = 1` to `/etc/security/pwquality.conf`.

Rule 5.3.3.2.7: Ensure password quality checking is enforced

- **Status:** FAIL
- **Evidence:** The `enforcing` option for `pam_pwquality` is not configured in `/etc/security/pwquality.conf` or inline PAM settings (`enforcing: null`).
- **Recommendation:** Configure `pam_pwquality` to enforce password quality checks. Add `enforcing = 1` to `/etc/security/pwquality.conf`.

Rule 5.3.3.2.8: Ensure password quality is enforced for the root user

- **Status:** FAIL

- **Evidence:** The `enforce_for_root` option for `pam_pwquality` is not configured in `/etc/security/pwquality.conf` or inline PAM settings (`enforce_for_root: null`).
- **Recommendation:** Configure `pam_pwquality` to enforce password quality for the root user. Add `enforce_for_root` to `/etc/security/pwquality.conf`.

Rule 5.3.3.3.1: Ensure password history remember is configured

- **Status:** FAIL
- **Evidence:** The `remember` option for `pam_pwhistory` is not configured in inline PAM settings (`remember: null`).
- **Recommendation:** Configure `pam_pwhistory` to remember at least 24 previous passwords. Add `remember=24` to the `pam_pwhistory.so` line in `/etc/pam.d/common-password`.

Rule 5.3.3.3.2: Ensure password history is enforced for the root user

- **Status:** FAIL
- **Evidence:** The `enforce_for_root` option for `pam_pwhistory` is not configured in inline PAM settings (`enforce_for_root: null`).
- **Recommendation:** Configure `pam_pwhistory` to enforce password history for the root user. Add `enforce_for_root` to the `pam_pwhistory.so` line in `/etc/pam.d/common-password`.

Rule 5.3.3.3.3: Ensure pam_pwhistory includes use_authtok

- **Status:** FAIL
- **Evidence:** The `use_authtok` option for `pam_pwhistory` is not configured in inline PAM settings (`use_authtok: null`).
- **Recommendation:** Configure `pam_pwhistory` to use `use_authtok`. Add `use_authtok` to the `pam_pwhistory.so` line in `/etc/pam.d/common-password`.

Rule 5.3.3.4.1: Ensure pam_unix does not include nullok

- **Status:** FAIL
- **Evidence:** The `pam_unix.so` module in `/etc/pam.d/common-auth` includes the `nullok` option (`pam_unix_flag_presence.nullok: "present"`).
- **Recommendation:** Remove the `nullok` option from the `pam_unix.so` line in `/etc/pam.d/common-auth`.

Rule 5.3.3.4.2: Ensure pam_unix does not include remember

- **Status:** PASS
- **Evidence:** The `pam_unix.so` module does not include the `remember` option (`pam_unix_flag_presence.remember: null`).
- **Recommendation:** None.

Rule 5.3.3.4.3: Ensure pam_unix includes a strong password hashing algorithm

- **Status:** PASS
- **Evidence:** The `pam_unix.so` module in `/etc/pam.d/common-password` includes

the `yescrypt` option (`pam_unix_flag_presence.yescrypt: "present"`), which is a strong hashing algorithm.

- **Recommendation:** None.

Rule 5.3.3.4.4: Ensure `pam_unix` includes `use_authtok`

- **Status:** FAIL
- **Evidence:** The `pam_unix.so` module does not include the `use_authtok` option (`pam_unix_flag_presence.use_authtok: null`).
- **Recommendation:** Add the `use_authtok` option to the `pam_unix.so` line in `/etc/pam.d/common-password`.

Section 5.4 – User Accounts and Environment

Rule 5.4.1.1: Ensure password expiration is configured

- **Status:** FAIL
- **Evidence:**
 - `PASS_MAX_DAYS` in `/etc/login.defs` is set to 99999.
 - Multiple user accounts (e.g., `root`, `daemon`, `sebti`) have `pass_max_days` set to 99999, indicating passwords do not expire.
- **Recommendation:** Set `PASS_MAX_DAYS` to 90 or less in `/etc/login.defs`. For existing users, update their password expiration using `sudo chage -M 90 <username>`.

Rule 5.4.1.2: Ensure minimum password days is configured

- **Status:** INFORMATIONAL
- **Evidence:** `PASS_MIN_DAYS` in `/etc/login.defs` is set to 0.
- **Recommendation:** None. CIS recommends setting `PASS_MIN_DAYS` to 1 or more to prevent users from changing their password multiple times in a short period to bypass history requirements.

Rule 5.4.1.3: Ensure password expiration warning days is configured

- **Status:** PASS
- **Evidence:** `PASS_WARN_AGE` in `/etc/login.defs` is set to 7.
- **Recommendation:** None.

Rule 5.4.1.4: Ensure strong password hashing algorithm is configured

- **Status:** PASS
- **Evidence:** `ENCRYPT_METHOD` in `/etc/login.defs` is set to `SHA512`, and `pam_unix` is configured to use `yescrypt`.
- **Recommendation:** None.

Rule 5.4.1.5: Ensure inactive password lock is configured

- **Status:** FAIL
- **Evidence:** `useradd_inactive` is set to `-1`, meaning accounts are never locked due to

inactivity.

- **Recommendation:** Set INACTIVE to 30 or less in `/etc/login.defs` to lock accounts after a period of inactivity. For existing users, update their inactive period using `sudo chage -I 30 <username>`.

Rule 5.4.1.6: Ensure all users last password change date is in the past

- **Status:** FAIL
- **Evidence:** Several user accounts (e.g., `usbmux`, `sebti`, `wazuh-indexer`, `wazuh`, `wazuh-dashboard`, `sshd`, `ftp`) have their `last_password_change` date set in the future (e.g., "Apr 19, 2026", "Jun 25, 2026").
- **Recommendation:** Force these users to change their passwords immediately or correct the `last_password_change` date to a past date using `sudo chage -d $(date +%Y-%m-%d) <username>`.

Rule 5.4.2.1: Ensure root is the only UID 0 account

- **Status:** PASS
- **Evidence:** Only `root` is listed as a UID 0 account (`uid_0_accounts: ["root"]`).
- **Recommendation:** None.

Rule 5.4.2.2: Ensure root is the only GID 0 account

- **Status:** PASS
- **Evidence:** Only `root` is listed as a primary GID 0 account (`primary_gid_0_accounts: ["root"]`).
- **Recommendation:** None.

Rule 5.4.2.3: Ensure group root is the only GID 0 group

- **Status:** PASS
- **Evidence:** Only `root` is listed as a GID 0 group (`gid_0_groups: ["root"]`).
- **Recommendation:** None.

Rule 5.4.2.4: Ensure root account access is controlled

- **Status:** FAIL
- **Evidence:** The `/etc/securetty` file does not exist (`securetty_exists: false`). This file is used to restrict root logins to specific TTYs.
- **Recommendation:** Create `/etc/securetty` and list only the TTYs from which the root user is allowed to log in directly.

Rule 5.4.2.5: Ensure root path integrity

- **Status:** UNKNOWN
- **Evidence:** The root user's `PATH` is `/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin`. The profile does not provide permissions for all directories in this path to verify integrity.
- **Recommendation:** Manually verify that all directories in the root user's `PATH` are owned by `root`, are not world-writable, and have appropriate permissions (e.g., `0755`).

Rule 5.4.2.6: Ensure root user umask is configured

- **Status:** FAIL
- **Evidence:** The `UMASK` setting in `/etc/login.defs` is `022` (`root_umask./etc/login.defs: "022"`). CIS recommends a more restrictive umask of `027` for the root user.
- **Recommendation:** Set the root user's umask to `027` by modifying the `UMASK` entry in `/etc/login.defs` to `027`.

Rule 5.4.2.7: Ensure system accounts do not have a valid login shell

- **Status:** FAIL
- **Evidence:** The `sync` system account has `/bin/sync` as its login shell, which is a valid shell.
- **Recommendation:** Change the login shell for the `sync` user to `/usr/sbin/nologin` or `/bin/false`: `sudo usermod -s /usr/sbin/nologin sync`.

Rule 5.4.2.8: Ensure accounts without a valid login shell are locked

- **Status:** PASS
- **Evidence:** All system accounts listed with `/usr/sbin/nologin` or `/bin/false` as their shell have a `passwd_status` of `L` (locked).
- **Recommendation:** None.

Rule 5.4.3.1: Ensure nologin is not listed in /etc/shells

- **Status:** PASS
- **Evidence:** `/usr/sbin/nologin` is not listed in `/etc/shells` (`nologin_in_shells: false`).
- **Recommendation:** None.

Rule 5.4.3.2: Ensure default user shell timeout is configured

- **Status:** FAIL
- **Evidence:** No `TMOUT` settings were found in `/etc/profile`, `/etc/profile.d/`, or `/etc/bash.bashrc` (`tmout_settings` are empty).
- **Recommendation:** Add `TMOUT=600` (or a lower value) to `/etc/profile.d/timeout.sh` to automatically log out idle users after 10 minutes.

Rule 5.4.3.3: Ensure default user umask is configured

- **Status:** FAIL
- **Evidence:** The `UMASK` setting in `/etc/login.defs` is `022` (`default_user_umask.login_defs: "022"`). CIS recommends a more restrictive umask of `027` for default users.
- **Recommendation:** Set the default user umask to `027` by modifying the `UMASK` entry in `/etc/login.defs` to `027`.

Section 6.1 – System Logging

Rule 6.1.1.1.1: Ensure journald service is active

- **Status:** PASS
- **Evidence:** The `systemd-journald` service is active (`systemd-journald_active: "active"`).
- **Recommendation:** None.

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use

- **Status:** PASS
- **Evidence:** The `systemd-journal-remote` package is not installed (`systemd-journal-remote_installed: false`).
- **Recommendation:** None.

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog

- **Status:** FAIL
- **Evidence:** The `ForwardToSyslog` option in `/etc/systemd/journald.conf` is commented out, implying the default value of `no` (`ForwardToSyslog=no`).
- **Recommendation:** Uncomment and set `ForwardToSyslog=yes` in `/etc/systemd/journald.conf` to ensure logs are forwarded to rsyslog. Restart `systemd-journald` and `rsyslog` services.

Rule 6.1.1.1.4: Ensure journald log file access is configured

- **Status:** INFORMATIONAL
- **Evidence:**
 - `/run/log/journal` directory mode: 2755, owner root, group 999.
 - `/var/log/journal` directory mode: 2755, owner root, group 999.
 - Log files within `/var/log/journal/...` have mode 0640, owner root, group 999.
- **Recommendation:** None. Review the permissions of `journald` log directories and files to ensure they meet organizational requirements.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured

- **Status:** INFORMATIONAL
- **Evidence:** The `journald.conf` file has `SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, `SystemMaxFiles`, `MaxRetentionSec`, and `MaxFileSec` commented out, indicating default values are in use.
- **Recommendation:** None. Manually review and configure `SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, `SystemMaxFiles`, `MaxRetentionSec`, and `MaxFileSec` in `/etc/systemd/journald.conf` to meet specific log retention requirements.

Rule 6.1.1.1.6: Ensure journald Storage is configured

- **Status:** FAIL
- **Evidence:** The `Storage` option in `/etc/systemd/journald.conf` is commented out, implying the default value of `auto`. CIS recommends `persistent`.
- **Recommendation:** Uncomment and set `Storage=persistent` in `/etc/systemd/journald.conf` to ensure journal logs are stored persistently across reboots. Restart

systemd-journald.

Rule 6.1.1.1.7: Ensure journald Compress is configured

- **Status:** PASS
- **Evidence:** The `Compress` option in `/etc/systemd/journald.conf` is commented out, implying the default value of `yes`.
- **Recommendation:** None.

Rule 6.1.2.1: Ensure rsyslog is installed

- **Status:** PASS
- **Evidence:** `rsyslog` package is installed (`rsyslog_installed: true`).
- **Recommendation:** None.

Rule 6.1.2.2: Ensure rsyslog service is enabled and active

- **Status:** PASS
- **Evidence:** `rsyslog.service` is enabled (`rsyslog_enabled: "enabled"`) and active (`rsyslog_active: "active"`).
- **Recommendation:** None.

Rule 6.1.2.3: Ensure rsyslog log file creation mode is configured

- **Status:** PASS
- **Evidence:** The `$FileCreateMode` directive in `/etc/rsyslog.conf` is set to `0640`.
- **Recommendation:** None.

Rule 6.1.2.4: Ensure rsyslog logging is configured

- **Status:** INFORMATIONAL
- **Evidence:** The `/etc/rsyslog.d/50-default.conf` file contains standard logging rules for various facilities.
- **Recommendation:** None. Manually review `/etc/rsyslog.conf` and files in `/etc/rsyslog.d/` to ensure all necessary system events are logged according to organizational policy.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log host

- **Status:** INFORMATIONAL
- **Evidence:** The `/etc/rsyslog.conf` file has remote forwarding directives (`module(load="imudp"), input(type="imudp" port="514"), module(load="imtcp"), input(type="imtcp" port="514")`) commented out, indicating no remote log host is configured.
- **Recommendation:** None. If remote logging is required, configure `rsyslog` to forward logs to a central log server.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client

- **Status:** PASS

- **Evidence:** The `imudp` and `imtcp` modules and their corresponding input directives are commented out in `/etc/rsyslog.conf`, indicating rsyslog is not configured to receive logs from remote clients.
- **Recommendation:** None.

Rule 6.1.2.7: Ensure logrotate is configured

- **Status:** INFORMATIONAL
- **Evidence:** `logrotate` is configured with a weekly rotation, 4 weeks retention, and includes configurations from `/etc/logrotate.d/`. The `logrotate.timer` is enabled and active.
- **Recommendation:** None. Manually review `/etc/logrotate.conf` and files in `/etc/logrotate.d/` to ensure log rotation policies meet organizational requirements for all critical log files.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed

- **Status:** FAIL
- **Evidence:** The `rsyslog-gnutls` package is not installed (`rsyslog_gnutls_installed: false`).
- **Recommendation:** Install `rsyslog-gnutls`: `sudo apt install rsyslog-gnutls`.

Rule 6.1.2.9: Ensure rsyslog forwarding uses gtls

- **Status:** UNKNOWN
- **Evidence:** `rsyslog-gnutls` is not installed, so it cannot be determined if rsyslog forwarding uses gtls.
- **Recommendation:** Install `rsyslog-gnutls` and then configure rsyslog to use GTLS for remote log forwarding if required.

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured

- **Status:** INFORMATIONAL
- **Evidence:** The profile does not contain specific configuration lines for `StreamDriverAuthMode` or `StreamDriverCACert` in rsyslog configuration files.
- **Recommendation:** None. If remote logging with GTLS is used, ensure CA certificates are properly configured for authentication.

Rule 6.1.3.1: Ensure access to all logfiles has been configured

- **Status:** FAIL
- **Evidence:**
 - `/var/log/wtmp` has mode 0664 (world-readable). CIS recommends 0640.
 - `/var/log/lastlog` has mode 0664 (world-readable). CIS recommends 0640.
 - `/var/log/faillog` has mode 0644 (world-readable). CIS recommends 0640.
- **Recommendation:** Set more restrictive permissions on these log files:
 - `sudo chmod 0640 /var/log/wtmp`
 - `sudo chmod 0640 /var/log/lastlog`
 - `sudo chmod 0640 /var/log/faillog`

Section 6.2 – System Auditing (auditd)

Rule 6.2.1.1: Ensure auditd packages are installed

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.1.2: Ensure auditd service is enabled and active

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.1.3: Ensure auditing for processes that start prior to auditd is enabled

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.1.4: Ensure audit_backlog_limit is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.1: Ensure audit log storage size is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.2: Ensure audit logs are not automatically deleted

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.3: Ensure system is disabled when audit logs are full

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.2.4: Ensure system warns when audit logs are low on space

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.1: Ensure changes to system administration scope (sudoers) is collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.2: Ensure actions as another user are always logged

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.3: Ensure events that modify the sudo log file are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.4: Ensure events that modify date and time information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.5: Ensure events that modify sethostname and setdomainname are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.6: Ensure events that modify /etc/issue and /etc/issue.net are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and

enable the service.

Rule 6.2.3.7: Ensure events that modify /etc/hosts and /etc/hostname are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.8: Ensure events that modify /etc/network and /etc/networks are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.9: Ensure events that modify /etc/netplan are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.10: Ensure use of privileged commands are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.11: Ensure events that modify /etc/group information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.12: Ensure events that modify /etc/passwd information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.13: Ensure events that modify /etc/shadow and /etc/gshadow are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.14: Ensure events that modify /etc/security/opasswd are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.15: Ensure events that modify /etc/nsswitch.conf file are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.16: Ensure events that modify /etc/pam.conf and /etc/pam.d/ information are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.17: Ensure unsuccessful file access attempts are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.18: Ensure discretionary access control permission modification events are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.19: Ensure successful file system mounts are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: sudo apt install auditd. Then configure and enable the service.

Rule 6.2.3.20: Ensure session initiation information is collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.21: Ensure login and logout events are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.22: Ensure file deletion events by users are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.23: Ensure events that modify the system's Mandatory Access Controls are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.24: Ensure successful and unsuccessful attempts to use the chcon command are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.25: Ensure successful and unsuccessful attempts to use the setfacl command are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.26: Ensure successful and unsuccessful attempts to use the chacl command are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

enable the service.

Rule 6.2.3.27: Ensure successful and unsuccessful attempts to use the usermod command are collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.28: Ensure kernel module loading unloading and modification is collected

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.29: Ensure the audit configuration is immutable

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.3.30: Ensure the running and on disk configuration is the same

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.1: Ensure the audit log file directory mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.2: Ensure audit log files mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.3: Ensure audit log files owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.4: Ensure audit log files group owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.5: Ensure audit configuration files mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.6: Ensure audit configuration files owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.7: Ensure audit configuration files group owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.8: Ensure audit tools mode is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.9: Ensure audit tools owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Rule 6.2.4.10: Ensure audit tools group owner is configured

- **Status:** FAIL
- **Evidence:** The auditd package is not installed (auditd_installed: false).
- **Recommendation:** Install auditd: `sudo apt install auditd`. Then configure and enable the service.

Section 6.3 – Integrity Checking

Rule 6.3.1: Ensure AIDE is installed

- **Status:** FAIL
- **Evidence:** The AIDE package is not installed (`aide_installed: false`).
- **Recommendation:** Install AIDE: `sudo apt install aide aide-common`. Run `sudo aideinit` to create the initial database.

Rule 6.3.2: Ensure filesystem integrity is regularly checked

- **Status:** FAIL
- **Evidence:** The AIDE package is not installed (`aide_installed: false`).
- **Recommendation:** Install AIDE: `sudo apt install aide aide-common`. Configure a cron job or systemd timer to regularly check filesystem integrity using AIDE.

Rule 6.3.3: Ensure cryptographic mechanisms are used to protect the integrity of audit tools

- **Status:** FAIL
- **Evidence:** The AIDE package is not installed (`aide_installed: false`).
- **Recommendation:** Install AIDE: `sudo apt install aide aide-common`. Configure AIDE to monitor the integrity of audit tools and their configuration files using strong cryptographic hashes.

Section 7.1 – Configure system file and directory access

Rule 7.1.1: Ensure access to /etc/passwd is configured

- **Status:** PASS
- **Evidence:** The `/etc/passwd` file has mode 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.2: Ensure access to /etc/passwd- is configured

- **Status:** PASS
- **Evidence:** The `/etc/passwd-` file has mode 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.3: Ensure access to /etc/group is configured

- **Status:** PASS
- **Evidence:** The `/etc/group` file has mode 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.4: Ensure access to /etc/group- is configured

- **Status:** PASS
- **Evidence:** The `/etc/group-` file has mode 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.5: Ensure access to /etc/shadow is configured

- **Status:** PASS
- **Evidence:** The `/etc/shadow` file has mode 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.6: Ensure access to `/etc/shadow-` is configured

- **Status:** PASS
- **Evidence:** The `/etc/shadow-` file has mode 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.7: Ensure access to `/etc/gshadow` is configured

- **Status:** PASS
- **Evidence:** The `/etc/gshadow` file has mode 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.8: Ensure access to `/etc/gshadow-` is configured

- **Status:** PASS
- **Evidence:** The `/etc/gshadow-` file has mode 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.9: Ensure access to `/etc/shells` is configured

- **Status:** PASS
- **Evidence:** The `/etc/shells` file has mode 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.10: Ensure access to `/etc/security/opasswd` is configured

- **Status:** PASS
- **Evidence:** The `/etc/security/opasswd` file has mode 0600, owner root, group root.
- **Recommendation:** None.

Rule 7.1.11: Ensure world writable files and directories are secured

- **Status:** PASS
- **Evidence:** No world-writable files or directories without the sticky bit were found in the scanned mounts (`world_writable_files` and `world_writable_dirs_no_sticky` lists are empty).
- **Recommendation:** None.

Rule 7.1.12: Ensure no files or directories without an owner and a group exist

- **Status:** PASS
- **Evidence:** No unowned or ungrouped files or directories were found in the scanned mounts (`unowned_or_ungrouped` list is empty).
- **Recommendation:** None.

Rule 7.1.13: Ensure SUID and SGID files are reviewed

- **Status:** INFORMATIONAL
- **Evidence:** The following SUID/SGID files were found:
 - /usr/bin/umount (4755 root:root)
 - /usr/bin/chsh (4755 root:root)
 - /usr/bin/sudo (4755 root:root)
 - /usr/bin/su (4755 root:root)
 - /usr/bin/passwd (4755 root:root)
 - /usr/bin/ssh-agent (2755 root:_ssh)
 - /usr/bin/fusermount3 (4755 root:root)
 - /usr/bin/newgrp (4755 root:root)
 - /usr/bin/chage (2755 root:shadow)
 - /usr/bin/expiry (2755 root:shadow)
 - /usr/bin/gpasswd (4755 root:root)
 - /usr/bin/crontab (2755 root:crontab)
 - /usr/bin/mount (4755 root:root)
 - /usr/bin/chfn (4755 root:root)
 - /usr/lib/dbus-1.0/dbus-daemon-launch-helper (4754 root:messagebus)
 - /usr/lib/x86_64-linux-gnu/utempter/utempter (2755 root:utmp)
 - /usr/lib/openssh/ssh-keysign (4755 root:root)
 - /usr/lib/polkit-1/polkit-agent-helper-1 (4755 root:root)
 - /usr/sbin/pam_extrausers_chkpwd (2755 root:shadow)
 - /usr/sbin/unix_chkpwd (2755 root:shadow)
- **Recommendation:** None. Regularly review SUID/SGID files to ensure they are legitimate and necessary for system operation. Remove SUID/SGID bits from any files that do not require them.

Section 7.2 – Local User and Group Settings

Rule 7.2.1: Ensure accounts in /etc/passwd use shadowed passwords

- **Status:** PASS
- **Evidence:** The system is configured to use shadowed passwords, as indicated by the presence of /etc/shadow with appropriate permissions (7.1.5) and the password status of user accounts.
- **Recommendation:** None.

Rule 7.2.2: Ensure /etc/shadow password fields are not empty

- **Status:** PASS
- **Evidence:** The root_shadow_password_status is * (locked), and user_chage_info shows last_password_change dates for users, implying non-empty password fields.
- **Recommendation:** None.

Rule 7.2.3: Ensure all groups in /etc/passwd exist in /etc/group

- **Status:** PASS
- **Evidence:** The orphan_gids_in_passwd list is empty, indicating no GIDs in /etc/passwd are missing from /etc/group.

- **Recommendation:** None.

Rule 7.2.4: Ensure shadow group is empty

- **Status:** PASS
- **Evidence:** The `shadow_group_members` list is empty, indicating no users are members of the shadow group.
- **Recommendation:** None.

Rule 7.2.5: Ensure no duplicate UIDs exist

- **Status:** PASS
- **Evidence:** The `duplicate_uids` object is empty, indicating no duplicate UIDs were found.
- **Recommendation:** None.

Rule 7.2.6: Ensure no duplicate GIDs exist

- **Status:** PASS
- **Evidence:** The `duplicate_gids` object is empty, indicating no duplicate GIDs were found.
- **Recommendation:** None.

Rule 7.2.7: Ensure no duplicate user names exist

- **Status:** PASS
- **Evidence:** The `duplicate_usernames` object is empty, indicating no duplicate user names were found.
- **Recommendation:** None.

Rule 7.2.8: Ensure no duplicate group names exist

- **Status:** PASS
- **Evidence:** The `duplicate_groupnames` object is empty, indicating no duplicate group names were found.
- **Recommendation:** None.

Rule 7.2.9: Ensure local interactive user home directories are configured

- **Status:** PASS
- **Evidence:** The `home_directory_issues` list is empty, indicating no issues with home directory existence or permissions for interactive users.
- **Recommendation:** None.

Rule 7.2.10: Ensure local interactive user dot files access is configured

- **Status:** PASS
 - **Evidence:** The `dotfile_permission_issues` list is empty, indicating no issues with dot file permissions in interactive user home directories.
 - **Recommendation:** None.
-

SUID/SGID Triage Report

```
{
  "findings": [
    {
      "path": "/usr/bin/ssh-agent",
      "owner": "root",
      "group": "_ssh",
      "mode": "2755",
      "suid": false,
      "sgid": true,
      "risk": "medium",
      "reason": "The `ssh-agent` utility typically does not require SGI",
      "recommendation": "Investigate the reason for the SGID permission"
    }
  ],
  "summary": {
    "total_checked": 20,
    "suspicious_found": 1
  }
}
```